Open-Vocabulary High-Resolution Remote Sensing Image Semantic Segmentation

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***Abstract*— Open-vocabulary image semantic segmentation (OVS) seeks to segment images into semantic regions across an open set of categories. Existing OVS methods commonly depend on foundational vision–language models and utilize similarity computation to tackle OVS tasks. However, these approaches are predominantly tailored to natural images and struggle with the unique characteristics of high-resolution remote sensing images, such as rapidly changing orientations and significant scale variations. To tackle this dilemma, we propose the first OVS framework specifically designed for high-resolution remote sensing imagery, introducing a rotation-aggregative similarity computation module to enhance segmentation across varying orientations and a multiscale feature integration strategy to generate scale-aware semantic masks. Additionally, we establish the first open-sourced OVS benchmark for remote sensing, comprising four public datasets. Experiments demonstrate that our framework effectively addresses orientation and scale challenges, achieving state-of-the-art performance. All codes and datasets are available at https://github.com/caoql98/OVRS**

***Index Terms*— Open-vocabulary semantic segmentation, remote sensing image segmentation, scale variation, varying orientations.**

I. Introduction

SEMANTIC segmentation of remote sensing imagery is a crucial task aimed at the precise pixel-level classification of various elements within an image, which plays a pivotal role in numerous applications [3], [4], [5], [6]. The advent of deep learning has revolutionized this field, enabling fully supervised segmentation techniques [7], [8], [9] to achieve remarkable accuracy and efficiency. These methods have been effectively employed in diverse domains, including detailed land-use and land-cover mapping [10], [11], [12], comprehensive traffic monitoring systems [13], [14], [15], and the accurate extraction of infrastructure such as buildings and roads [15], [16], [17]. Despite their success, these fully supervised approaches are highly dependent on the availability of large-scale, well-annotated datasets [18], [19], [20]. However, this demand for extensive high-resolution remote sensing image datasets presents significant challenges, as collecting and annotating such datasets is both costly and time-intensive. More critically, models trained on these datasets tend to overfit the specific categories encountered during training, limiting their ability to generalize to new, unseen categories.

Transfer learning [21], [22], [23] with limited samples of unseen categories has been explored as a strategy for enhancing generalization in segmentation tasks. These approaches adapt pretrained models, which have learned rich representations from abundant annotated categories, to new categories with limited annotations. While this enables knowledge transfer from extensive data corpora, it still faces significant challenges when applied to unseen categories. The scarcity of training samples can lead to instability and ambiguity in the learned representations. Furthermore, some unseen categories may lack any annotated samples, with only category names available, further complicating the adaptation process.

Thus, as illustrated in Fig. 1(b), to tackle the challenge of segmenting unseen categories, many researchers have introduced zero-shot segmentation algorithms [24], [25], [26]. These algorithms rely solely on class names as text embeddings to infer the semantic regions for categories not present in the training datasets yet in a closed category set. Despite these advancements, such methods remain confined to individual datasets. Consequently, when deployed in real-world environments for high-resolution remote sensing images, where models frequently encounter objects beyond their closed category set, their performance often degrades sharply. This limitation stems from the inherent nature of zero-shot learning, which lacks explicit training samples for unseen categories and instead relies on learned representations that may not effectively capture the complexity of dynamic remote sensing scenes.

To overcome this limitation and extend the applicability of segmentation models to more realistic scenarios, open-vocabulary image semantic segmentation (OVS) [27], [28], [29] has been proposed, as depicted in Fig. 1(c). Unlike transfer learning, which adapts with limited annotated samples, or zero-shot segmentation, which infers unseen categories

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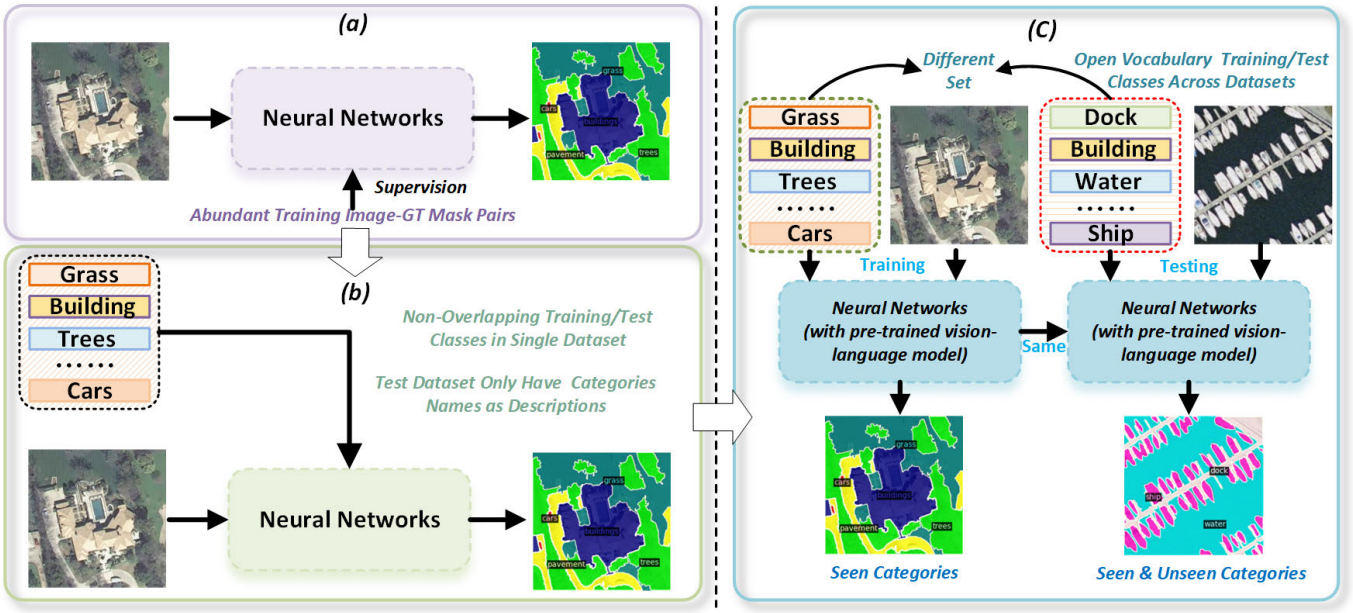
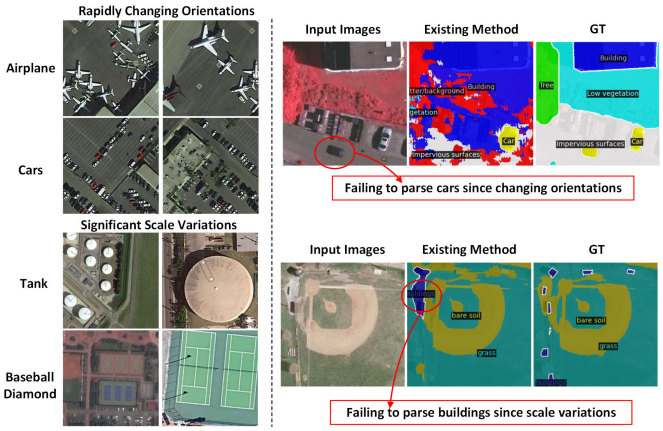


Fig. 1. Differences between diverse semantic segmentation tasks for remote sensing images. (a) Fully supervised segmentation required abundant annotated data. (b) Zero-shot segmentation requires only class names for test datasets in a closed set. (c) Open-vocabulary segmentation could achieve open-set segmentation across datasets with only category names.



(a) (b)

Fig. 2. (a) Challenges for remote sensing images including rapidly changing orientations and significant scale variations. (b) Corresponding failure examples for the specific remote sensing challenges. Existing methods: OVS performance with existing methods [1], [2]. GT: Ground truth.

from pre-existing embeddings, OVS aims to segment remote sensing images by considering an open set of categories across varying datasets. Existing OVS methods mainly utilize the generalized vision–language models like contrastive language-image pre-training (CLIP) [28] as the foundational models and utilize the similarities between the features of images and the textual embeddings of category names from the generalized foundational models to judge the semantic regions. For instance, Liang et al. [29] proposed two-stage OVS models, which first generate the mask proposals and subsequently segment the semantic regions based on the similarities of mask-category pairs. Similarly, Freeseg [30] adopted the two-stage framework and further introduced prompt learning for the text encoder, which helps the network perform the OVS tasks in the one-in-all pattern. Following the prompt learning manner, SegPrompt successfully leverages the category information as prompts to improve the model's class-agnostic segmentation ability for the OVS tasks. Though these OVS have achieved some success, they all focus on natural images, which could not handle the unique traits of high-resolution remote sensing imagery. As shown in Fig. 2(a), rapidly changing orientations and significant scale variations are general challenges in high-resolution remote sensing image understanding tasks, and many studies [31], [32], [33], [34], [35] address this issue by proposing specialized methods. As shown in Fig. 2(b), these problems would consistently hamper the OVS performance for high-resolution remote sensing imagery and seriously impede the development of open-vocabulary Earth-vision tasks.

To address the challenges inherent in open-vocabulary semantic segmentation of high-resolution remote sensing imagery, we introduce a novel algorithm designed to be the first OVS framework for high-resolution remote sensing imagery, named OVRS. One of the primary obstacles in remote sensing is the rapid change in object orientations, which poses a significant challenge for traditional segmentation methods. We posit that an effective solution requires a mechanism capable of adaptively capturing and aggregating semantic information across various orientations, thereby minimizing the impact of these variations. To this end, we propose a rotation-aggregative similarity computation module that synthesizes orientation-adaptive similarity maps to serve as the foundation for initial semantic maps. OVRS begins by rotating the input high-resolution remote sensing images across multiple orientations. These rotated images are then processed through the vision branch of a pretrained CLIP model to extract orientation-varying image features. Concurrently, the language branch of CLIP generates class-specific features for the target categories. By computing the similarity between

the orientation-specific image features and the class features, the network produces a set of orientation-varying similarity maps. These maps are subsequently aggregated across all orientations, yielding rotation-adaptive similarity maps that form the initial semantic maps. These initial maps are then subjected to further refinement at both spatial and categorical levels, enhancing their accuracy and robustness.

In addition to addressing orientation variability, OVRS tackles the significant scale variations typical of remote sensing imagery by leveraging features from multiple levels of the feature extraction network. Each level of the network captures image features at different scales, which are crucial for accurate segmentation. During the upsampling process, these multiscale features are progressively integrated into the semantic maps, enriching them with comprehensive scale-related information and resulting in the generation of scale-aware semantic maps. Specifically, at each stage of upsampling, the semantic maps are averaged in both channel and spatial dimensions to produce semantic activation vectors. These vectors are then used to selectively activate the image features, which are subsequently combined with the semantic maps to form progressively refined, scale-aware semantic representations. This multiscale integration strategy ensures that the final semantic maps are not only orientation-adaptive but also scale-aware, thus significantly improving the segmentation performance across diverse and dynamic remote sensing scenarios.

By addressing the unique challenges of high-resolution remote sensing imagery, we introduce a novel approach to open-vocabulary segmentation in Earth vision. The key contributions of this article are summarized as follows.

1) We propose the first OVS framework specifically designed for high-resolution remote sensing imagery, establishing a new research direction in Earth vision and enabling more adaptable segmentation models that generalize beyond closed-category datasets.
2) To address the challenges of rapidly changing orientations and significant scale variations, we introduce the rotation-aggregative similarity computation module and progressively generate scale-aware semantic maps through the integration of multiscale features, resulting in more accurate and robust segmentation.
3) We extend the applicability of vision–language models like CLIP to remote sensing OVS by introducing newly designed mechanisms that bridge the domain gap, enhancing the effectiveness of vision–language alignment in Earth observation tasks.

II. Related Work

In this section, we first review related work on remote sensing image semantic segmentation. Then, the advanced research with regard to open-vocabulary semantic segmentation is extensively introduced.

A. Remote Sensing Image Semantic Segmentation

Remote sensing image semantic segmentation focuses on the precise delineation of semantic regions within remote sensing imagery [31], [36]. The advent of deep learning has significantly advanced this field, leading to the development of numerous sophisticated algorithms. For example, Li et al. [37] developed efficient attention modules to capture contextual dependencies, enhancing segmentation accuracy. Building on this, end-to-end attention-based semantic segmentation network (SSAtNet) [38] introduced a pyramid attention module that leverages multiscale features for adaptive refinement of segmentation features. Prioritizing computational efficiency, local attention network (LANet) [39] introduced patch attention and local embeddings to achieve effective segmentation through a patch-focused approach, while multistage attention resiudal U-net (ResU-Net) [40] incorporated a linear attention mechanism to improve computational efficiency over traditional dot-product attention. Focusing on spatial information, high-resolution context extraction network (HRCNet) [41] developed a high-resolution context extraction network to better capture global contextual information, thereby enhancing segmentation performance. The Transformer architecture [42], known for its superior image understanding capabilities across various computer vision tasks, has also been adapted for remote sensing image segmentation. For instance, Xu et al. [43] adopt the efficient Transformer to perform the semantic segmentation task. Based on this concept, UNet-like Transformer (UNetFormer) [44] constructs the Transformer-based decoder and further proposes an UNetFormer for real-time remote sensing segmentation. Inspired by the powerful global modeling capabilities of the Swin Transformer, He et al. [45] adopt the Swin Transformer to mine pixel-level correlation to enhance the feature representation ability. To boost the global information extracted by the Transformer network with local information from the convolutional neural network (CNN), Zhang et al. [46] proposed a Transformer and CNN hybrid segmentation network, where the Transformer is adopted for long-range spatial dependencies modeling, and the CNN is utilized to maintain the local details. Similarly, fusing swin transformer and convolutional neural network (STransFuse) [47] jointly leveraged the Swin Transformer and CNN to extract coarse-grained and fine-grained feature representations and perform a staged semantic segmentation at diverse semantic scales.

B. Open-Vocabulary Semantic Segmentation

Open-vocabulary semantic segmentation seeks to accurately delineate semantic regions across an unrestricted set of categories. Current methods primarily utilize foundational vision–language models like CLIP [28], leveraging the similarities between textual category names and image features to identify corresponding semantic regions, as shown in Fig. 3(a). For instance, Liang et al. [29] fine-tune CLIP using paired masked image regions and text descriptions, enabling efficient classification of these masked regions. Based on the powerful generalization ability of CLIP, CLIPseg [48] and Freeseg [30] follow a prompt learning approach [49], [50], directly using text descriptions as prompts to segment query images. Building on this adapter-based approach, side adapter network (SAN) [51] introduces a side adapter network that generates mask proposals and attention biases, guiding the deeper layers of CLIP for proposal-wise classification. Similarly, SegCLIP [52] introduces the concept of super-pixels

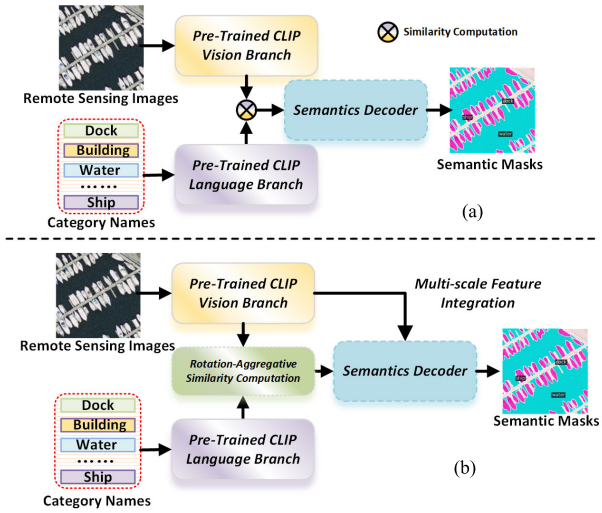


Fig. 3. (a) Other existing methods for natural images. (b) Proposed method (OVRS) for high-resolution remote sensing images.

into OVS, aggregating image patches around learnable centers to form semantic regions based on the generalized pretrained CLIP features. Extending this patch aggregation approach, Chen et al. [53] performed OVS by summarizing localized regions of the target image and distilling visual concepts from pretrained CLIP models. Focusing on the semantic alignment between visual content and unbounded text, semanticassisted calibration network (SCAN) [54] incorporates a generalized semantic prior and a contextual shift strategy to enhance segmentation performance. In contrast, simple encoder-decoder network (SED) [1] addresses the often-overlooked local spatial information by utilizing a CNN-based CLIP to construct an efficient OVS network. Focusing on efficiency, global knowledge calibration network (GKC) [55] performs the fast open-vocabulary segmentation through knowledge distillation on pretrained CLIP. More directly, cost aggregation segmentation network (CAT-Seg) [2] explores the multimodal nature established between image and text embeddings of the pretrained CLIP model, performing segmentation through cost volume computation. However, all these methods are constructed based on natural images and fail to address the unique characteristics of remote sensing imagery. To tackle this challenge and draw inspiration from these distinct traits, we propose the first open-vocabulary semantic segmentation framework specifically designed for remote sensing images. Remote sensing images have unique characteristics such as rapidly changing orientations and significant scale variations, which are not adequately addressed by existing OVS methods designed for natural images. As shown in Fig. 3(b), OVRS is specifically designed to handle these challenges, making it the first of its kind in the remote sensing domain. We perform extensive experiments on the newly proposed remote sensing OVS benchmark, demonstrating the state-of-the-art OVS performance for high-resolution remote sensing images.

III. Proposed Method

A. Method Overview

Consider a high-resolution remote sensing image I and a set of potential class categories $C = \{T(n)\}$, where $n = 1, \ldots, N_C$, with $T(n)$ representing the textual description of the nth category and N_C being the total number of classes. For instance, $T(n)$ could be defined as "an image of Category Name." The task of open-vocabulary semantic segmentation requires assigning a label to each pixel in the image I based on these categories. Unlike traditional semantic segmentation tasks, open-vocabulary segmentation faces the unique challenge of working with a flexible and dynamic set C provided as free-form text.

The CLIP model consists of a visual encoder and a text encoder that generate image embeddings and corresponding text embeddings, respectively. During training, the CLIP model maximizes the cosine similarity between the image and its matched text and minimizes the cosine similarity between the image and its unmatched text. The large-scale training image–text pairs allow the CLIP model to show powerful generalized recognition ability. Following previous methods [1], [2], [54], we employ the pretrained CLIP model as the foundational vision–language framework to leverage its generalized knowledge. The overall structure of OVRS is depicted in Fig. 4. Specifically, given query remote sensing images, they are first rotated through various orientations (e.g., 0°, 90°, 180°, and 270°) and then passed through the vision branch of the pretrained CLIP model to extract orientation-specific image features. Concurrently, category names are input into the pretrained CLIP language branch to generate corresponding class features. In the rotation-aggregative similarity computation module, the cosine similarities between the image and class features are computed for each orientation to produce orientation-specific similarity maps. These maps are aligned and aggregated across orientations to generate rotation-adaptive similarity maps, which serve as the foundation for initial semantic predictions. Subsequently, through, respectively, slicing these semantic maps at category and spatial dimensions, the Transformer layers are applied to these semantic maps to refine the prediction in both spatial and categorical refinement stages. To address significant scale variations, we integrate multiscale image features from different levels of the feature extraction network into the middle semantic maps during the upsampling process. Intermediate semantic maps are iteratively fused with multiscale features through spatial and channel activations, This feature fusion could ensure semantic details overlooked due to scale differences are recovered. Particularly, the multiscale features are first average pooled in spatial and channel dimensions, and the averaged vectors are leveraged to activate the middle maps to provide the neglected details. This process produces final scale-aware semantic masks that are both orientation-adaptive and scale-sensitive. The proposed method effectively combines the strengths of rotation-aggregative similarity computation and multiscale integration, enabling robust semantic segmentation tailored to the unique challenges of remote sensing imagery.

B. Rotation-Aggregative Similarity Computation

Objects in high-resolution remote sensing images tend to exist in varying orientations. Existing OVS methods often neglect this trait and fail to tackle the varying orientations. Previous work [31] has demonstrated the importance of

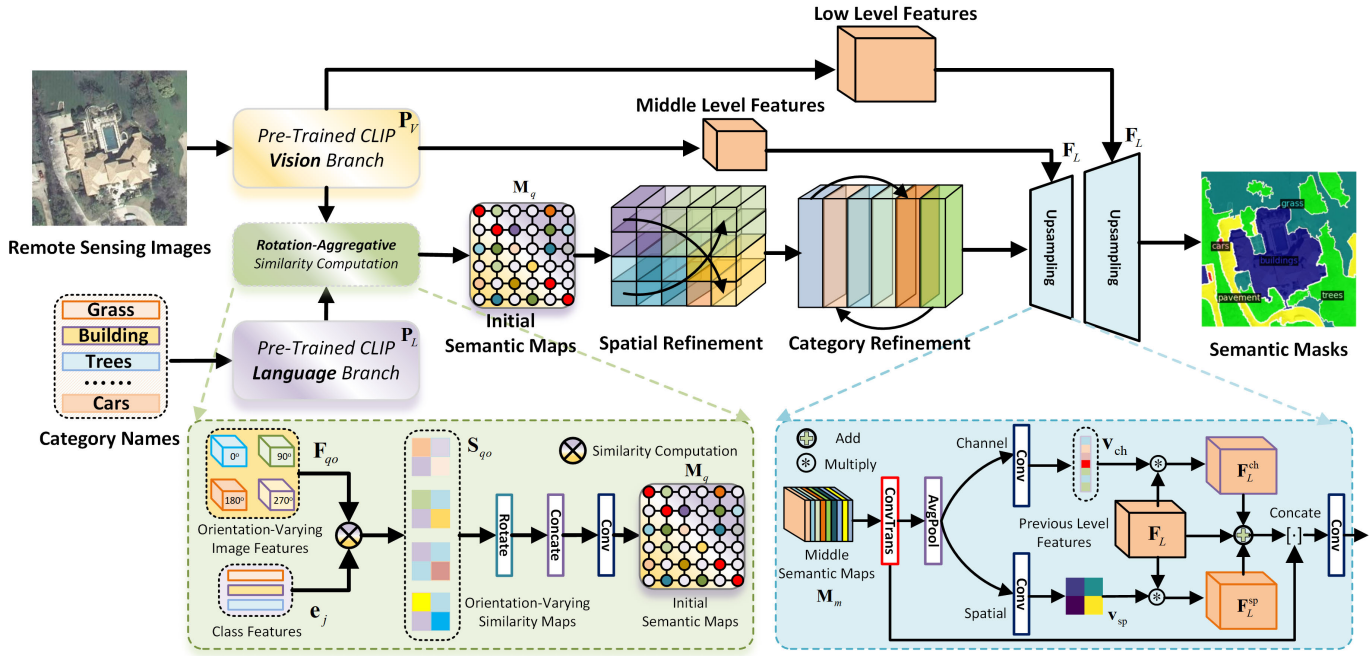


Fig. 4. Overall framework of the proposed open-vocabulary high-resolution remote sensing image semantic segmentation. Query images are initially rotated at multiple angles to generate orientation-specific image features using the vision branch of CLIP as the feature extractor. Simultaneously, category names are passed through the language branch to derive text embeddings as class features. By performing rotation-aggregative similarity computations between the orientation-specific image features and class features, the initial semantic maps are generated, capturing orientation-adaptive semantics. These maps are further refined iteratively in both spatial and categorical manner to enhance their precision. Additionally, to address scale variations, features from different levels are integrated during the upsampling process to progressively refine the semantic maps, leading to the final scale-aware semantic masks.

rotation-invariant methods for remote sensing segmentation by using rotation-adaptive matching in a rotation-aggregative manner. Drawing inspiration from this. Thus, we argue that the semantics from varying orientations should be gathered to ensure objects with varying orientations can be accurately parsed. Particularly, given the query remote sensing image $\mathbf{i}_{q1}$, the image is first rotated in diverse orientations, for instance, 90°, 180°, and 270°, to acquire orientation-varying images $\{\mathbf{I}_{q1}, \mathbf{I}_{q2}, \ldots, \mathbf{I}_{qN_A}\}$, where N_A denotes the number of orientations. These images are further propagated into the frozen vision branch $\mathbf{P}_V$ of pretrained CLIP to obtain the diverse image features from different angles $\{\mathbf{f}_{q1}, \mathbf{f}_{q2}, \ldots, \mathbf{f}_{qN_A}\} \in \mathbb{R}^{H\times W\times d}$

$$\mathbf{f}_{qi} = \mathbf{P}_V(\mathbf{I}_{qi}), \quad i \in \{1, 2, \ldots, N_A\}. \tag{1}$$

These image features are concatenated in a single dimension to obtain the orientation-varying image features $\mathbf{F}_{qo} \in \mathbb{R}^{HW\times N_A\times d}$

$$\mathbf{F}_{qo} = \left[\mathbf{f}_{q1}, \mathbf{f}_{q2}, \ldots, \mathbf{f}_{qN_A}\right]. \tag{2}$$

Meanwhile, the category names $\{C_1, C_2, \ldots, C_{N_C}\}$ of target semantic objects working as the textual descriptions like "an image of category name" are inputted into the frozen language branch $\mathbf{P}_L$ of pretrained CLIP to acquire the corresponding text embeddings as the class features $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_{N_C}\} \in \mathbb{R}^{1\times d}$

$$\mathbf{e}_j = \mathbf{P}_L(C_j), \quad j \in \{1, 2, \ldots, N_C\}. \tag{3}$$

Cosine similarity is widely utilized in OVS methods, particularly those employing foundational vision–language models such as CLIP [28]. CLIP uses cosine similarity as a key component for contrastive learning, and as a result, OVS methods [1], [2] typically leverage cosine similarity to infer the semantics of each pixel corresponding to a particular category. Thus, the orientation-varying semantics are obtained through the cosine similarities between class features and the orientation-varying image features

$$\mathbf{S}_{qo}[:, :, j] = \frac{\mathbf{F}_{qo}\cdot\mathbf{e}_j}{\|\mathbf{F}_{qo}\|\|\mathbf{e}_j\|} \tag{4}$$

where $\mathbf{S}_{qo} \in \mathbb{R}^{HW\times N_A\times N_C}$ denotes the corresponding semantic maps. For better subsequent semantic understanding, these maps are processed along the category dimension with a single convolutional layer

$$\mathbf{S}'_{qo}[:, :, j] = \mathrm{Conv}(\mathbf{S}_{qo}[:, :, j]) \tag{5}$$

where $\mathbf{S}'_{qo} \in \mathbb{R}^{HW\times N_A\times N_C\times d_F}$. Then, the maps are rotated along the angle dimension N_A to the original orientation of $\mathbf{I}_{q1}$ and further fused in the angle dimension to rotation-aggregately obtain the initial semantic maps $\mathbf{M}_q \in \mathbb{R}^{HW\times N_C\times d_F}$

$$\mathbf{M}_q = \mathrm{Conv}\left(\left[\mathbf{S}'_{qo}[:, 0, :, :], \ldots, \mathrm{Rotate}(\mathbf{S}'_{qo}[:, N_A, :, :])\right]\right). \tag{6}$$

In this manner, the generated semantic maps adaptively aggregate semantic information across various orientations and successfully handle the rapidly changing orientations.

C. Spatial and Category Refinement

The acquired initial semantic maps $\mathbf{M}_q$ are the coarse perceptions of query remote sensing images within the image–text

semantic space of the pretrained CLIP. To refine this coarse result, the semantic maps should be better analyzed, respectively, in both vision and text modalities.

First, to refine the holistic vision understanding of the semantic maps, the semantic maps are sliced at the category dimension N_C, which helps the network concentrate on the vision level and perform the spatial refinement at the pixel level. We adopt the Swin Transformer [56] as the refinement layer. More specifically, the refinement could be defined as follows:

$$\mathbf{M}'_q[:, i, :] = L_T^{\text{sp}}\big(\mathbf{M}_q[:, i, :]\big), \quad i \in \{1, 2, \ldots, N_C\} \tag{7}$$

where $\mathbf{M}_q[:, i, :] \in \mathbb{R}^{(H\times W)\times d_F}$, and L_T^{sp} denotes a pair of two consecutive Swin Transformer layers for spatial refinement. Notably, d_F is the channel dimension for each token, and attention is performed on individual categories, respectively.

Furthermore, the semantic maps are refined at the text modality and the category refinement helps the network to precisely capture the relationships between multiple categories. Correspondingly, the semantic maps $\mathbf{M}'_q$ are sliced at the spatial dimension HW to eliminate the effect of vision modality. We also adopt the Swin Transformer as the category refinement layer, and the process is defined as follows:

$$\mathbf{M}''_q[i, :, :] = L_T^{\text{ca}}\big(\mathbf{M}'_q[i, :, :]\big), \quad i \in \{1, 2, \ldots, HW\} \tag{8}$$

where $\mathbf{M}'_q[i, :, :] \in \mathbb{R}^{N_C\times d_F}$. Different from the spatial refinement, there exist no spatial relations between categories, thus a linear Transformer layer L_T^{ca} without position embeddings is adopted for the category refinement. The progressive refinement process is a well-established paradigm in segmentation tasks [57], [58], [59], used to generate more precise segmentation results. This paradigm is, therefore, incorporated in OVRS for improved accuracy. The overall refinement including both spatial and category refinements would repeat several times like CAT-Seg [2] to acquire the middle semantic maps with higher accuracy.

D. Scale-Aware Upsampling

Although the rotation-aggregative similarity computation effectively enables the network to handle rapidly changing orientations, and the spatial and category refinement generates more precise semantic maps, significant scale variations remain unaddressed. To tackle this issue, it is crucial to incorporate multiscale image features from the feature extractor to enhance the model's scale adaptation ability since these multiscale features have different focusing object scales. Specifically, since we use the pretrained CLIP vision branch as the feature extractor, the image features from various layers of this branch are collected to provide vital scale information. The inclusion of these multiscale features ensures that the intermediate semantic maps capture previously neglected details, making the model more robust to scale variations.

Particularly, in a single upsampling process, given the middle semantic maps $\mathbf{M}_m \in \mathbb{R}^{HW\times N_C\times d_F}$, and the previous level features $\mathbf{F}_L \in \mathbb{R}^{d_F\times H_L\times W_L}$, the middle semantic maps are first average pooled in spatial and channel levels, and then processed through convolutional layers to obtain spatial and channel activation vectors

$$\mathbf{v}_{\text{sp}} = \text{Conv}\big(\text{AvgPool}_{\text{spatial}}(\text{Conv}(\mathbf{M}_m))\big) \tag{9}$$

$$\mathbf{v}_{\text{ch}} = \text{Conv}\big(\text{AvgPool}_{\text{channel}}(\text{Conv}(\mathbf{M}_m))\big) \tag{10}$$

where $\mathbf{v}_{\text{sp}} \in \mathbb{R}^{HW\times 1}$ and $\mathbf{v}_{\text{ch}} \in \mathbb{R}^{d_F\times 1}$. These vectors contain the neglected scale details, respectively, in spatial and channel levels. Through the dot-product $\odot$ between these vectors and the previous level features $\mathbf{F}_L$, the relevant semantic regions are activated

$$\mathbf{F}_L^{\text{sp}} = \mathbf{v}_{\text{sp}} \odot \mathbf{F}_L \tag{11}$$

$$\mathbf{F}_L^{\text{ch}} = \mathbf{v}_{\text{ch}} \odot \mathbf{F}_L. \tag{12}$$

The aggregated activation features are then concatenated with the middle semantic maps, incorporating scale information to produce scale-aware middle semantic maps

$$\mathbf{M}_m^s = \text{Conv}\big(\big[\mathbf{F}_L^{\text{sp}} + \mathbf{F}_L^{\text{ch}} + \mathbf{F}_L, \mathbf{M}_m\big]\big). \tag{13}$$

Finally, the upsampling process is repeated iteratively, ensuring that the semantic maps recover both fine-grained spatial details and scale information. The final output semantic masks are supervised by cross-entropy loss

$$\mathcal{L} = -\sum \mathbf{M}_{\text{GT}} \log(\mathbf{M}_F) \tag{14}$$

where $\mathbf{M}_{\text{GT}}$ and $\mathbf{M}_F$ denote the ground truth and predicted masks, respectively.

To summarize the proposed open-vocabulary semantic segmentation method for remote sensing imagery, we present a concise mathematical formulation. The process can be represented as

$$\mathbf{M}_F = \mathcal{U}\Big(\mathcal{R}\Big(\text{Concat}\big(\mathcal{S}\big(\mathbf{P}_V\big(\mathbf{I}_q^{(r)}\big), \mathbf{P}_L(\mathbf{C})\big)\big)_{r=1}^{N_A}, \mathcal{F}_{\text{multi}}\Big)\Big) \tag{15}$$

where the components are defined as follows.

1) $\mathbf{I}_q^{(r)}$: Rotated versions of the query image $\mathbf{I}_q$, where $r \in \{1, 2, \ldots, N_A\}$ corresponds to N_A different orientations (e.g., 0°, 90°, 180°, and 270°).
2) $\mathbf{P}_V$: Vision branch of the pretrained CLIP model, used to extract image features.
3) $\mathbf{P}_L$: Language branch of the pretrained CLIP model, used to extract text embeddings for the open-vocabulary class names $\mathbf{C}$.
4) $\mathcal{S}(\mathbf{F}, \mathbf{E})$: Similarity computation (e.g., cosine similarity) between the image features $\mathbf{F}$ and class embeddings $\mathbf{E}$, generating semantic maps.
5) $Concat(\cdot)$: Concatenation of semantic maps across N_A orientations.
6) $\mathcal{R}(\cdot)$: Rotation-aggregative similarity fusion to align and aggregate orientation-specific semantic maps.
7) $\mathcal{F}_{multi} = \{\mathbf{F}^{(l)}\}_{l=1}^{L}$: Multiscale features extracted from L layers of the feature extraction network ($\mathbf{P}_V$).
8) $\mathcal{U}(\cdot, \cdot)$: Scale-aware upsampling, incorporating multiscale features ($\mathcal{F}_{\text{multi}}$) to refine and reconstruct high-resolution segmentation masks.
9) $\mathbf{M}_F$: Final segmentation mask, generated by fusing orientation-aware and scale-aware semantic information.

Algorithm 1 Open-Vocabulary Semantic Segmentation (OVS) for Remote Sensing Images

Input: Query image $\mathbf{I}_q$, categories names $\{C_1, C_2, \ldots, C_{N_C}\}$, CLIP branches $\mathbf{P}_V, \mathbf{P}_L$, orientations N_A, Swin Transformers $L_T^{\text{sp}}, L_T^{\text{ca}}$, multiscale features $\mathbf{F}_L$
Output: Final segmentation mask $\mathbf{M}_F$
{Rotation-Aggregative Similarity Computation}
for each orientation $r = 1, 2, \ldots, N_A$ **do**
  Rotate $\mathbf{I}_q$ to get $\mathbf{I}_q^{(r)}$
  Extract features $\mathbf{f}_{q_r} = \mathbf{P}_V(\mathbf{I}_q^{(r)})$ with frozen vision branch of pretrained CLIP
end for
Concatenate features: $\mathbf{F}_{qo} = [\mathbf{f}_{q_1}, \mathbf{f}_{q_2}, \ldots, \mathbf{f}_{q_{N_A}}]$
{Generate Class Features}
for each category $j = 1, 2, \ldots, N_C$ **do**
  Generate class feature $\mathbf{e}_j = \mathbf{P}_L(C_j)$ with frozen language branch of pretrained CLIP
end for
{Compute Similarities}
for each category $j = 1, 2, \ldots, N_C$ **do**
  $\mathbf{S}_{qo}[:, :, j] = \frac{\mathbf{F}_{qo} \cdot \mathbf{e}_j}{\|\mathbf{F}_{qo}\| \|\mathbf{e}_j\|}$
end for
{Spatial and Category Refinement (4 iterations)}
for each category $j = 1, 2, \ldots, N_C$ **do**
  for iteration $k = 1, 2, 3, 4$ **do**
    Apply spatial refinement: $\mathbf{M}'_q[:, i, :] = L_T^{\text{sp}}(\mathbf{M}_q[:, i, :])$
    Apply category refinement: $\mathbf{M}''_q[h, :, :] = L_T^{\text{ca}}(\mathbf{M}'_q[h, :, :])$
  end for
end for
{Scale-Aware Upsampling}
Compute activation vectors $\mathbf{v}_{\text{sp}}, \mathbf{v}_{\text{ch}}$
Compute activated features $\mathbf{F}_L^{\text{sp}}, \mathbf{F}_L^{\text{ch}}$
Concatenate and fuse features: $\mathbf{M}_m^s = \text{Conv}([\mathbf{F}_L^{\text{sp}} + \mathbf{F}_L^{\text{ch}} + \mathbf{F}_L, \mathbf{M}_m])$
{Final Output}
Return $\mathbf{M}_F = \mathbf{M}_m^s$

This formulation encapsulates the entire process, starting from the input query image $\mathbf{I}_q$ and class names $\mathbf{C}$, through feature extraction, similarity computation, rotation-aggregative fusion, multiscale integration, and scale-aware upsampling, to produce the final semantic segmentation mask $\mathbf{M}_F$. To facilitate better analysis of the entire process, we provide the pseudocode in Algorithm 1.

IV. Experiments

To thoroughly evaluate the effectiveness of the proposed method, we conduct an extensive set of experiments on the newly introduced open-vocabulary remote sensing semantic segmentation benchmark. The experimental section is organized as follows. First, we provide a detailed description of the datasets used and the evaluation metrics adopted. Next, we outline the key implementation details of OVRS. Following this, a comprehensive analysis of the performance comparison between OVRS and state-of-the-art OVS approaches is presented, both in qualitative and quantitative terms. Finally, we perform a series of ablation studies to assess the contribution of each component of OVRS.

A. Dataset and Evaluation Metric

To comprehensively evaluate the proposed method, we expand the experimental setup beyond the commonly used Potsdam and Vaihingen semantic segmentation datasets by including the processed dense labeling remote sensing dataset (DLRSD) [61] and instance segmentation in aerial images datasets (iSAID) [60] datasets, thereby establishing a more robust OVS benchmark for remote sensing imagery. We follow the few-shot segmentation benchmark [60] to preprocess these remote sensing datasets. The processed DLRSD dataset consists of 7002 images across 17 categories, while the processed iSAID dataset includes 24 439 images across 15 categories. Potsdam and Vaihingen, which share the same six categories, contain 20 102 and 2254 images, respectively. Detailed category names for these datasets are provided in Table I. The Potsdam and Vaihingen datasets share the same categories, which might introduce some evaluation biases. However, as these datasets are collected from different regions, such biases are alleviated to some extent.

In our experimental setup, DLRSD and iSAID are used as the training datasets, while evaluations are conducted across all datasets to assess the OVS performance. Specifically, 5601 images from the DLRSD dataset and 1401 images from iSAID are used for training, with the remaining 1401 and 6363 images used for validation, respectively. Following standard evaluation protocols from prior OVS methods [1], [2], we utilize mean intersection over union (mIoU), frequency weighted intersection over union (fwIoU), and mean accuracy (mACC) as evaluation metrics, providing a comprehensive reflection of the model's performance. For further real-world applications, commonly used and meaningful categories in remote sensing could be selected as training categories, which could potentially improve the generalization ability of the open-vocabulary segmentation method and boost the parsing performance on unseen categories during inference.

The mIoU is computed as the average IoU across all classes

$$\text{IoU}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i}, \quad \text{mIoU} = \frac{1}{N} \sum_{i=1}^{N} \text{IoU}_i \tag{16}$$

where TP_i, FP_i, and FN_i represent true positives, false positives, and false negatives for class i, and N is the total number of classes.

The fwIoU metric accounts for the frequency of each class in the dataset

$$\text{fwIoU} = \frac{1}{\sum_{i=1}^{N} n_i} \sum_{i=1}^{N} n_i \cdot \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \tag{17}$$

where n_i is the number of pixels in class i.

Finally, mACC measures the per-class accuracy, defined as the ratio of correctly classified pixels to the total number of pixels for each class

$$\text{ACC}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i}, \quad \text{mACC} = \frac{1}{N} \sum_{i=1}^{N} \text{ACC}_i. \tag{18}$$

TABLE I
CATEGORIES FOR iSAID, DLRSD, POTSDAM, AND VAIHINGEN DATASETS

Dataset	Category Names
iSAID [60]	ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, bridge, large vehicle, small vehicle, helicopter, swimming pool, roundabout, soccer ball field, plane, harbor
DLRSD [61]	airplane, bare soil, buildings, cars, chaparral, court, dock, field, grass, mobile home, pavement, sand, sea, ship, tanks, trees, water
Potsdam [62]	impervious surfaces, Building, Low vegetation, Tree, Car, background
Vaihingen [63]	impervious surfaces, Building, Low vegetation, Tree, Car, background

TABLE II
PERFORMANCE COMPARISONS OF DIFFERENT METHODS WITH DIFFERENT DATASETS AS THE TRAINING DATASETS. VAL DATA DENOTES THE VALIDATION DATASET OF THE CHOSEN TRAINING DATASET

Method	Val Data			DLRSD			iSAID			Potsdam			Vaihingen		
	mIoU	fwIoU	mACC	mIoU	fwIoU	mACC	mIoU	fwIoU	mACC	mIoU	fwIoU	mACC	mIoU	fwIoU	mACC
DLRSD as training dataset.															
SCAN [54]	57.82	66.38	75.01	48.52	55.01	68.68	34.18	44.62	49.05	20.22	28.37	34.70	5.38	7.72	22.54
SAN [51]	83.28	88.32	89.55	85.73	86.24	91.03	30.63	39.87	44.03	**30.30**	**38.04**	**44.98**	31.92	40.23	45.36
FreeSeg [30]	89.64	91.67	92.81	85.46	86.39	91.39	30.13	40.14	41.49	20.87	28.16	18.57	28.82	29.77	43.37
GKC [55]	90.24	91.71	94.08	85.24	86.09	90.54	29.95	39.86	39.98	22.47	27.16	38.67	30.18	36.61	46.37
SED [1]	90.43	92.03	94.48	85.13	86.80	91.36	21.54	25.72	36.28	19.47	21.29	33.40	29.40	36.77	49.38
CAT-Seg [2]	90.94	93.07	95.22	85.84	86.79	91.44	23.56	27.40	38.48	26.79	31.28	44.72	32.32	38.71	49.65
OVRS	**91.11**	**93.22**	**95.45**	**85.98**	**86.94**	**91.52**	**39.09**	**48.97**	**54.43**	27.47	26.81	42.07	**33.71**	**38.57**	**50.01**
iSAID as training dataset.															
SCAN [54]	64.59	70.15	76.73	17.09	19.50	41.11	60.37	62.41	74.43	19.41	25.91	40.17	9.88	13.16	26.40
SAN [51]	75.72	82.72	84.62	20.28	19.20	**47.95**	85.67	88.02	90.99	14.79	17.69	36.51	16.38	17.53	35.38
FreeSeg [30]	76.13	82.89	85.13	20.68	20.79	43.81	91.81	91.68	94.31	17.67	21.23	32.87	13.45	12.81	28.45
GKC [55]	78.37	85.01	88.16	24.75	23.89	43.97	91.89	91.93	94.87	16.48	20.16	31.12	13.03	10.31	32.68
SED [1]	77.25	84.70	87.97	**25.45**	24.09	44.54	92.92	92.16	95.49	15.27	18.96	28.57	12.82	9.61	29.37
CAT-Seg [2]	80.78	86.64	89.01	22.13	23.93	44.21	92.33	92.51	95.19	14.68	14.40	37.47	9.93	10.87	38.40
OVRS	**87.87**	**89.07**	**92.30**	20.67	**26.18**	45.44	**93.29**	**92.52**	**95.77**	**19.94**	**26.29**	**49.12**	**20.82**	**26.61**	**50.60**

TABLE III
MEAN PERFORMANCE COMPARISONS OF DIFFERENT METHODS. THE BEST RESULTS ARE SHOWN IN BOLD

Method	DLRSD for training			iSAID for training		
	mIoU	fwIoU	mACC	mIoU	fwIoU	mACC
SCAN [54]	33.22	40.42	49.99	34.26	38.22	51.77
SAN [51]	52.37	58.54	62.99	42.57	45.03	59.09
FreeSeg [30]	50.98	55.23	57.53	43.95	45.88	56.91
GKC [55]	51.62	56.29	61.93	44.90	46.26	58.16
SED [1]	49.19	52.52	60.98	44.74	45.90	57.19
CAT-Seg [2]	51.89	55.45	63.90	43.97	45.67	60.85
OVRS	**55.47**	**58.90**	**66.70**	**48.52**	**52.13**	**66.65**

B. Implementation Details

We use the pretrained CLIP vision transformer base network (ViTB)/16 [64] as the foundational vision–language feature extractor, leveraging its robust visual and textual embeddings. The method is implemented using PyTorch and Detectron2 frameworks, ensuring flexibility and efficiency for large-scale experimentation. For optimization, we employ the Adam optimizer with decoupled weight decay (AdamW) optimizer [65] with an initial learning rate of $2e^{-4}$ to balance convergence speed and stability, alongside a weight decay of $10e^{-4}$ to mitigate overfitting. Training is conducted with a batch size of 4 on an Nvidia A100 GPU with 80 GB of memory, which is crucial for handling high-resolution remote sensing imagery. All images are resized to $3 \times 384 \times 384$ to ensure consistency across datasets and compatibility with the CLIP model. The models are trained for 100 000 iterations, allowing sufficient convergence for learning complex features necessary for open-vocabulary semantic segmentation. After training on the DLRSD/iSAID datasets, we evaluate the models on both their validation sets and a comprehensive benchmark consisting of four remote sensing datasets (Potsdam, Vaihingen, DLRSD, and iSAID) to assess their generalization performance.

C. Performance Analysis

The quantitative performance comparisons between OVRS and other advanced OVS algorithms are detailed in Table II, with the highest results highlighted in bold. When trained on DLRSD, OVRS outperforms others on most metrics. Notably, compared with the state-of-the-art CAT-Seg method, OVRS shows substantial improvements on the iSAID dataset, with an increase of 15.53% in mIoU, 11.57% in fwIoU, and 15.95% in mACC. Across the validation datasets of DLRSD, Potsdam,

TABLE IV
PERFORMANCE COMPARISONS OF DIVERSE CLASSES ON THE VAIHINGEN DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	impe. sur.	Building	Low vegetation	Tree	Car	background	mean IoU
DLRSD as the training dataset							
SCAN [54]	13.69	13.93	0.86	0.23	2.51	1.06	5.38
SAN [51]	**42.59**	58.50	14.75	41.50	34.03	0.15	31.92
FreeSeg [30]	37.19	54.75	13.41	37.18	28.23	2.13	28.82
GKC [55]	39.09	55.13	14.75	38.06	32.19	1.89	30.18
SED [1]	41.29	50.75	**15.36**	36.40	29.43	3.15	29.40
CAT-Seg [2]	37.54	65.22	9.20	38.05	36.71	7.19	32.32
OVRS	37.27	**66.07**	2.50	**42.35**	**44.77**	**9.29**	**33.71**
iSAID as the training dataset							
SCAN [54]	14.54	15.92	6.81	14.93	5.87	1.24	9.88
SAN [51]	21.65	0.62	**21.34**	28.22	25.17	1.29	16.88
FreeSeg [30]	27.13	6.75	2.13	21.23	21.75	1.75	13.45
GKC [55]	25.46	10.75	4.76	19.40	16.75	1.09	13.03
SED [1]	**28.79**	0.30	4.48	0.12	**41.26**	1.95	12.82
CAT-Seg [2]	26.28	1.40	2.83	10.17	15.99	0.93	9.93
OVRS	21.79	**36.69**	2.80	**43.85**	16.23	**3.56**	**20.82**

and Vaihingen, OVRS also demonstrates superior performance with consistent and significant gains. However, it falls short of the top performance on the Potsdam dataset compared to the SAN method, though it still ranks second in mIoU. This underperformance may be attributed to a potential dataset domain gap between the DLRSD dataset and the Potsdam dataset, which could hamper the segmentation performance on Potsdam. Addressing this dataset domain gap remains a challenge for future work.

In contrast, when trained on the larger iSAID dataset, OVRS achieves the best results across nearly all metrics. Most notably, the Vaihingen dataset sees dramatic performance gains, with mIoU and fwIoU both surpassing 20%, while the mACC of OVRS reaches 50.60%. These results underscore the generalization capability of OVRS when given a diverse and extensive dataset for training. Nevertheless, on the DLRSD dataset, OVRS fails to surpass all metrics, specifically mIoU and mACC, suggesting the persistent gap between DLRSD and iSAID data distributions, which reflects the inherent variability in remote sensing data sources.

To further understand the performance differences, we also provide average performance comparisons in Table III. Notably, open-vocabulary remote sensing (OVR) outperforms others in all metrics, regardless of whether DLRSD or iSAID is used for training. Interestingly, with a relatively small DLRSD as the training dataset, OVRS still achieves the best average results though fails in Potsdam, indicating that DLRSD images may offer more representative features for diverse categories. In particular, the highest metric improvements are seen in mACC, where a 2.80% boost was achieved, bringing the overall mACC to 66.70%. These results reaffirm the effectiveness of OVRS, emphasizing its capability to leverage the unique traits of remote sensing imagery. This significant performance improvement further highlights the necessity of tailoring OVS algorithms to the specific challenges of Earth observation, setting a new benchmark for OVS in remote sensing.

To delve deeper into the performance, a detailed breakdown of class-wise performance comparisons on the Vaihingen dataset is presented in Table IV. Overall, OVRS demonstrates superior performance, achieving 33.71% mIoU when trained on DLRSD and 20.82% mIoU with iSAID as the training dataset. Specifically, using DLRSD for training, OVRS shows the best results on the building class with a remarkable 66.07% mIoU, while the most significant performance gain is observed on the car class, showing an improvement of approximately 8% mIoU. When trained on the iSAID dataset, OVRS performs exceptionally well on the tree class, with a leading 43.85% mIoU, and achieves the most substantial performance boost on the building class, showing a striking near 21% mIoU improvement. One intriguing observation is that although all OVS methods struggle with segmenting pixels annotated as background, OVRS still manages to show improvements in this area. This suggests that the incorporation of remote sensing-specific traits meaningfully contributes to better semantic understanding in Earth observation tasks. These improvements underscore the ability of OVRS to adapt to the unique challenges of remote sensing imagery and successfully push the boundaries of open-vocabulary segmentation in this domain. Despite these successes, OVRS fails to precisely parse the low vegetation category. We believe the relatively low performance of the low vegetation class is primarily due to two reasons. First, the term “low vegetation” is an inherently vague description with a broad range of meanings, which introduces ambiguity in semantic understanding. Second, the semantic similarity between “low vegetation” and other categories, such as “tree,” makes it challenging for the model to precisely distinguish between these classes in an open-vocabulary manner. In the future, we aim to design networks with higher discriminative capabilities to better handle such ambiguous categories.

To further assess the effectiveness of the proposed method, we visualize the segmentation results in Fig. 5, with the critical

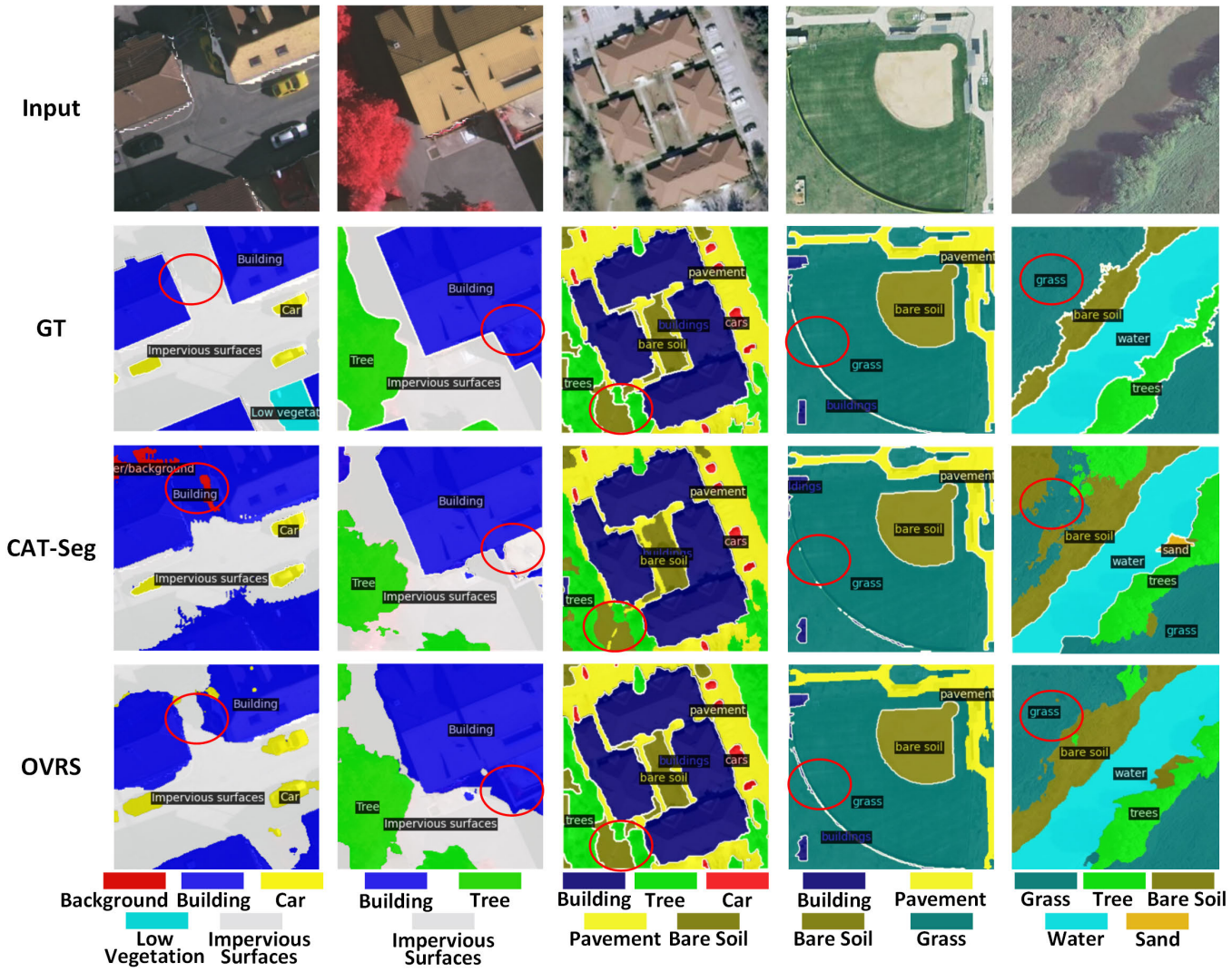


Fig. 5. Qualitative results of the proposed method. (From top to bottom) Input images, the ground truth of query images, predictions of CAT-Seg, and predictions of OVRS. The red circle highlights the critical difference.

differences highlighted by the red circle. OVRS outperforms existing approaches by accurately segmenting regions and enhancing prediction quality. Notably, the rotation-aggregative similarity computation module significantly improves the model's ability to distinguish remote sensing objects, reducing misclassification. For instance, in the first and second columns, OVRS successfully classifies more pixels as impervious surfaces and demonstrates greater precision in identifying building areas compared to the CAT-Seg model.

Moreover, the scale-aware upsampling process proves to be a key factor in enhancing semantic detail parsing. In the third and fourth columns, where other methods wrongly classify detailed pixels as pavement or grass, OVRS successfully corrects these misclassifications, identifying them as bare soil and buildings, respectively. These qualitative results highlight the importance of incorporating unique traits of remote sensing imagery, enabling our model to achieve more precise semantic segmentation.

As illustrated in Fig. 6, a particularly noteworthy observation is the ability of OVRS to rectify annotation errors present in the ground truth. For instance, in the first row, several pixels erroneously labeled as pavement in the ground truth are accurately classified by OVRS as grass and sand, a correction

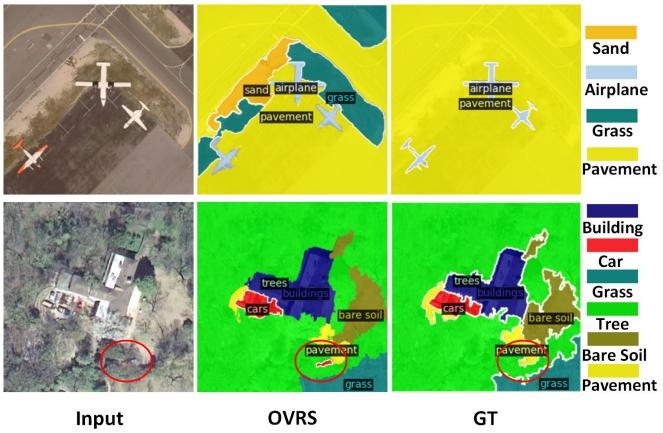


Fig. 6. Some intriguing visualization results of OVRS. The red circle denotes the annotation errors.

TABLE V

ABLATION STUDY OF THE ROTATION-AGGREGATIVE SIMILARITY COMPUTATION AND SCALE-AWARE UPSAMPLING. RSC ILLUSTRATES THE ROTATION-AGGREGATIVE SIMILARITY COMPUTATION, AND SAU DENOTES THE SCALE-AWARE UPSAMPLING

RSC	SAU	mIoU	fwIoU	mACC
		80.78	86.65	89.02
✓		87.06	89.03	91.22
	✓	85.81	88.47	90.64
✓	✓	**87.87**	**89.07**	**92.30**

TABLE VI

ABLATION STUDY OF DIVERSE ORIENTATIONS

Orientations	mIoU	fwIoU	mACC
[0°]	80.78	86.65	89.02
[0°,90°]	85.79	87.79	90.93
[0°,90°,180°]	86.18	88.13	91.09
[0°,90°,180°,270°]	**87.06**	**89.03**	**91.22**
[0°,45°,...,270°,315°]	86.86	88.63	90.91

TABLE VII

ABLATION STUDY OF THE LAYER NUMBER OF THE SCALE-AWARE UPSAMPLING PROCESS

Layer Number	mIOU	fwIoU	mACC
0	80.78	86.65	89.02
1	84.66	87.89	90.17
2	**85.81**	**88.47**	**90.64**
3	85.62	88.11	90.23

TABLE VIII

GENERALIZATION PERFORMANCE COMPARISON (MIOU) BETWEEN TRADITIONAL REMOTE SENSING SEGMENTATION METHODS AND OVRS

Methods	DLRSD	Potsdam	Vaihingen
DeepLabv3 [66]	2.58	8.41	6.98
Segformer [67]	2.72	8.76	7.01
ST-UNet [45]	2.88	8.63	7.83
FTransUNet [68]	3.02	8.78	7.69
OVRS	**20.67**	**19.94**	**20.82**

that is visually validated by the input image. This demonstrates the capacity of the model to refine spatial labeling even in the presence of erroneous annotations. Similarly, in the second row, OVRS reclassifies pixels annotated as pavement in the ground truth as cars, a segmentation outcome consistent with the visual evidence from the input image.

These findings emphasize the genuine open-vocabulary capabilities of OVRS, which extends beyond reliance on predefined category lists typically constrained by the training data. The model demonstrates the ability to recognize and appropriately label novel categories that were neither explicitly annotated nor represented in the training process. This capability has profound implications for remote sensing tasks, particularly in dynamic environments where flexibility and adaptability to unforeseen category distributions are critical. Such outcomes not only validate the robustness of OVRS but also highlight its potential to advance the accuracy and applicability of semantic segmentation in complex and diverse real-world scenarios.

D. Ablation Study

To assess the effectiveness of the proposed OVS framework for high-resolution remote sensing imagery, we conduct a series of experiments to analyze the impact of key components. For a fair comparison, we use the same foundational vision–language models as other OVS methods, with CAT-Seg serving as the baseline. All ablation experiments are performed using the iSAID dataset for training and its corresponding validation set for evaluation. We examine the effects of rotation-aggregative similarity computation, scale-aware upsampling, diverse orientations, and the number of layers in the scale-aware upsampling process. The evaluation metrics include mIoU, fwIoU, and mACC.

First, we analyze the contributions of the rotation-aggregative similarity computation and the scale-aware upsampling process individually, with results summarized in Table V. Notably, the network's ability to handle varying orientations significantly improves segmentation performance, boosting the mIoU from 80.78% to 87.06%. Addressing scale variations also leads to substantial gains, resulting in a performance increase of approximately 5%. By jointly tackling these unique remote sensing traits, OVRS achieves a new state-of-the-art performance. A deeper analysis reveals that handling orientation variations provides greater performance benefits compared to scale variations, which indicates that both orientations and scales are critical factors contributing to the final segmentation performance, yet the parsing results benefit more from the orientations.

In the OVRS, we rotate input images at three angles (90°, 180°, and 270°) along with the original image to account for orientation-varying objects. To study the impact of aggregating multiple orientations in the rotation-aggregative similarity computation module, the experimental results are shown in Table VI. As expected, increasing the number of orientations improves segmentation performance incrementally. A significant performance gain is observed when introducing the first new orientation, further reinforcing the importance of handling rapidly changing orientations. The network achieves peak performance when aggregating 0°, 90°, 180°, and 270°. Beyond this, adding more orientations slightly hampers performance, likely due to redundant information and the increased parameter count, which could interfere with the extraction of rotation-adaptive features. Therefore, the combination of these four primary orientations appears to strike the optimal balance, functioning effectively as the four axes of an orientation system for generating rotation-aggregative features.

Moreover, as shown in Table VII, we conduct experiments to analyze the number of layers from which features are introduced to address the rapidly changing scale variations. Sequentially incorporating features from earlier layers improves segmentation performance. Specifically, the best results are achieved when features from two previous layers are included. However, incorporating features from additional

TABLE IX
EFFICIENCY COMPARISONS BETWEEN DIVERSE SOTA METHODS

Methods	Params. (M)	GFLOPs	mIoU
SCAN [54]	593.2	857.2	64.59
SAN [51]	8.4	64.3	75.72
FreeSeg [30]	349.2	1537.6	76.13
GKC [55]	40.5	151.4	78.37
SED [1]	90.7	254.3	77.25
CAT-Seg [2]	433.7	2121.4	80.78
OVRS	447.5	2626.5	87.87

shallower layers slightly degrades performance. This suggests that excessively shallow features may lack sufficient scale information and contain overly coarse semantics, which is not beneficial for the upsampling process in remote sensing imagery. Consequently, introducing features from two layers strikes the optimal balance between providing scale-aware details and maintaining semantic coherence for accurate segmentation.

Overall, the ablation studies clearly demonstrate that both the rotation-aggregative similarity computation and the scale-aware upsampling process are essential to the success of the proposed method. By addressing the unique orientation and scale challenges inherent to remote sensing imagery, the method significantly enhances segmentation performance, achieving state-of-the-art results. Additionally, the careful selection of orientation aggregation and feature layers proves crucial in optimizing the network's ability to adapt to these challenges, ensuring more accurate and robust semantic segmentation in open-vocabulary settings. These insights not only validate the importance of the proposed components but also provide a foundation for future improvements in remote sensing OVS.

OVRS is particularly beneficial in actual applications where new or previously unseen classes are likely to appear, such as disaster monitoring, environmental surveys, and land-use analysis. These scenarios often involve dynamically updating or expanding category sets based on evolving needs, and the names of high-concerned objects can be obtained. By leveraging open-vocabulary segmentation, OVRS provides flexibility in handling unseen classes without requiring exhaustive prior annotations, making it highly applicable in dynamic and large-scale remote sensing tasks.

V. DISCUSSION

To assess the generalization capability of OVRS, we trained several traditional remote sensing segmentation models on the iSAID dataset and tested them on other datasets. The results, measured in terms of mIoU, are shown in Table VIII. These findings demonstrate that OVRS outperforms traditional segmentation techniques in terms of generalization performance. This advantage can be attributed to OVRS's reliance on the robust generalization capabilities of foundational vision–language models, enabling open-vocabulary segmentation. In contrast, traditional methods remain heavily dependent on extensive pixel-level annotated data within a closed set, limiting their adaptability to unseen categories. Moreover, to assess the efficiency of OVRS, we compare it with state of the art (SOTA) approaches in terms of parameter count (Params.), floating-point operations (GFLOPs), and mIoU performance on the validation set, using iSAID as the training dataset (see Table IX). While OVRS incurs additional computational overhead due to the rotation-aggregative similarity computation and scale-aware upsampling, this tradeoff yields promising performance gains. These results highlight the balance between efficiency and effectiveness, underscoring the viability of OVRS for open-vocabulary segmentation in remote sensing. Our proposed open-vocabulary semantic segmentation framework for high-resolution remote sensing imagery effectively addresses challenges related to rapidly changing orientations and significant scale variations. By leveraging a rotation-aggregative similarity computation module and multiscale feature integration, OVRS enhances segmentation accuracy under open-vocabulary settings. However, remote sensing imagery presents additional complexities, such as spectral discrepancies between datasets and variations in spatial resolution, which can influence model generalization. While OVRS primarily focuses on geometric and scale-related challenges, future research could explore spectral-aware feature representations and domain adaptation techniques to improve robustness across more diverse datasets. Moreover, integrating foundation models trained on multisensor remote sensing data may further enhance adaptability and performance. This study provides a strong foundation for advancing OVS in remote sensing and encourages future efforts to address broader challenges in Earth vision tasks.

VI. CONCLUSION

In this article, we introduced the first OVS benchmark for high-resolution remote sensing imagery and developed a novel open-sourced OVS framework specifically tailored for Earth vision. To address the unique challenges of high-resolution remote sensing imagery, such as rapidly changing orientations and significant scale variations, we proposed a rotation-aggregative similarity computation module that generates orientation-adaptive similarity maps. These maps are subsequently refined at spatial and categorical levels to enhance segmentation accuracy. Additionally, we integrated multiscale image features into the upsampling process to produce scale-aware semantic masks, enabling the framework to handle significant scale variations. Extensive experiments on four public high-resolution remote sensing datasets under open-vocabulary settings demonstrated that OVRS achieves state-of-the-art performance, affirming its effectiveness in Earth vision tasks. Despite these promising results, the OVRS method still faces challenges related to computational burden. Additionally, it does not account for spectral differences between datasets or variations in resolution. In future work, we aim to explore model lightweight techniques to enhance computational efficiency. Furthermore, spectral-aware feature representations and domain adaptation methods could be incorporated to mitigate spectral discrepancies between datasets and address variations in spatial resolution.

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